islands and continents on earth. Some entrepreneurs hope eventually to explore and conquer new planets and even galaxies, but in the meantime, the modern economy has had to find a better method of expanding.

Science has provided modernity with the alternative. The fox economy cannot grow, because foxes don't know how to produce more rabbits. The rabbit economy stagnates, because rabbits cannot make the grass grow faster. But the human economy can grow because humans can discover new materials and sources of energy.

The traditional view of the world as a pie of a fixed size presupposes there are only two kinds of resources in the world: raw materials and energy. But in truth, there are three kinds of resources: raw materials, energy and knowledge. Raw materials and energy are exhaustible – the more you use, the less you have. Knowledge, in contrast, is a growing resource – the more you use, the more you have. Indeed, when you increase your stock of knowledge, it can give you more raw materials and energy as well. If I invest $100 million searching for oil in Alaska and I find it, then I now have more oil, but my grandchildren will have less of it. In contrast, if I invest $100 million researching solar energy, and I find a new and more efficient way of harnessing it, then both I and my grandchildren will have more energy.

For thousands of years, the scientific road to growth was blocked because people believed that holy scriptures and ancient traditions already contained all the important knowledge the world had to offer. A corporation that believed all the oil fields in the world had already been discovered would not waste time and money searching for oil. Similarly, a human culture that believed it already knew everything worth knowing would not bother searching for new knowledge. This was the position of most premodern human civilisations. However, the Scientific Revolution freed humankind from this conviction. The greatest scientific discovery was the discovery of ignorance. Once humans realised how little they knew about the world, they suddenly had a very good reason to seek new knowledge, which opened up the scientific road to progress.